

A square-root obstruction to uniform endpoint-profile rates

Research proof; independent mathematical review completed

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Scope. We first give a counterexample in the general finite, positively weighted monotone comparison-tensor class, then transfer its robust inequality to completely finite equiprobable dice. It concerns the rate of agreement of individual columns, rather than the already sharp additive offset in the one-sided relative bound.

Theorem 1. *For all sufficiently large m there is a finite monotone tensor $p_i(q) \in [0, 1]$, $1 \leq i \leq m$, with positive row weights w_q summing to one and*

$$\sum_q w_q \prod_{i \in S} p_i(q) = \frac{1}{|S| + 1} \quad (S \subseteq [m]),$$

having a row with $1 \leq s := \sum_i (1 - p_i) \leq 100$ at which

$$\max_i |m(1 - p_i) - s| \geq cm^{-1/2}.$$

All but one of the columns are identical. Thus a universal endpoint profile estimate $o(m^{-1/2})$ on bounded s is impossible in this tensor class.

Lemma 1. *Let $Q = \lceil (m + 1)/2 \rceil$. The Q -node Gauss–Legendre probability rule on $[0, 1]$ has a node a and weight v with*

$$m(1 - a) \longrightarrow \pi^2, \quad v \geq cm^{-3/2}.$$

Proof. Number the nodes downwards from 1, and choose index $j = \lfloor \sqrt{m} \rfloor$. Write $2a - 1 = \cos \theta$. The elementary zero bounds in DLMF 18.16.2, specialized at Jacobi parameters 0, 0, give

$$\frac{(j - 1/2)\pi}{Q + 1/2} \leq \theta \leq \frac{j\pi}{Q + 1/2}.$$

Thus $\theta \sim 2\pi/\sqrt{m}$. Since $1 - a = (1 - \cos \theta)/2$, the asserted location follows.

For the weight, no quadrature-weight asymptotic is needed. The Bernstein inequality for Legendre polynomials is

$$(1 - x^2)^{1/4} |P_r(x)| \leq \sqrt{2/\pi} (r + 1/2)^{-1/2} \quad (-1 < x < 1).$$

This is DLMF 18.14.7 with Gegenbauer parameter $1/2$ (also 18.14.3 at Jacobi parameters 0, 0). Hence the shifted Legendre reproducing kernel satisfies

$$K_{Q-1}(a, a) = \sum_{r=0}^{Q-1} (2r + 1) P_r(2a - 1)^2 \leq \frac{CQ}{\sin \theta}.$$

At a Gaussian node its probability weight is $v = K_{Q-1}(a, a)^{-1}$. To verify the identity, the kernel polynomial at that node, divided by its diagonal value, is the cardinal Lagrange polynomial there; integrate its square using exactness through degree $2Q - 2$ and the reproducing identity. Thus $v \geq c \sin \theta / Q \geq cm^{-3/2}$. $\square$

Proof of Theorem 1. Let $\sigma = \sum_{\ell=1}^Q v_\ell \delta_{a_\ell}$ be the Gaussian rule, and F its cumulative distribution function. It integrates every polynomial of degree at most m , because $2Q - 1 \geq m$. On the continuous row parameter $t \in [0, 1]$, with uniform row measure, set

$$p_1(t) = F(t), \quad p_2(t) = \cdots = p_m(t) = t.$$

All columns are nondecreasing. A mixed moment not involving column 1 is immediate. If it involves column 1 and r other columns, $0 \leq r \leq m - 1$, then Fubini gives

$$\int_0^1 F(t) t^r dt = \frac{1 - \int a^{r+1} d\sigma(a)}{r+1} = \frac{1}{r+2}.$$

Thus all required mixed moments hold.

At the node a of Lemma 1, the two one-sided values of F differ by v . At least one, denoted F_* , satisfies $|F_* - a| \geq v/2$. The corresponding comparison row has

$$s = m(1 - a) + (a - F_*), \quad m(1 - p_1) - s = (m - 1)(a - F_*).$$

The first lies in $[1, 100]$ for large m , since $|a - F_*| \leq 1$; the second has absolute value at least $cm^{-1/2}$.

Finally make the row measure finite without losing this row. Partition $[0, 1]$ at all nodes a_ℓ . On each resulting interval, F is constant, so every mixed-moment integrand above is a polynomial of degree at most m . Apply a positive Gauss–Lobatto quadrature of degree at least m separately on every interval, retaining both endpoints. At a shared endpoint keep the two resulting rows distinct, with the left and right values of F , respectively. Their other $m - 1$ coordinates agree, and ordering the lower F value first preserves monotonicity. The positive weights sum to one and preserve every mixed moment. Both one-sided comparison rows at a occur with positive weight, so the required one is retained. $\square$

Corollary 1 (Actual finite uniform dice). *For all sufficiently large m there is a permutation-fair family of $m + 1$ finite uniform dice and a face of one die whose comparison probabilities satisfy*

$$1 \leq s = \sum_{i=1}^m (1 - p_i) \leq 100, \quad \max_i |m(1 - p_i) - s| \geq cm^{-1/2}.$$

No bound on the face counts is asserted here.

Proof. Use the Gaussian measure σ , node a , weight v , and favorable one-sided CDF value F_* in the preceding proof. Choose $t \in (0, 1)$ sufficiently close to a on that side, and strictly between Gaussian nodes, so that $|t - a| < v/8$ and $F_\sigma(t) = F_*$. Also require $m|t - a| < 1/10$; this is possible by arbitrarily close approach. Then

$$|F_\sigma(t) - t| \geq 3v/8, \quad m(1 - t) \longrightarrow \pi^2.$$

(For the displayed limit one may choose $m|t - a| \rightarrow 0$.) Apply the realization theorem in `finite_profile_realization.tex` with $\rho < v/100$ and, if necessary, still smaller so that $m\rho < 1/10$. For the resulting comparison row, put $e_i = p_i - t$ for $i \geq 2$. Then

$$m(1 - p_1) - s = (m - 1)(t - p_1) + \sum_{i=2}^m e_i.$$

Its absolute value is at least $(m - 1)(3v/8 - 2\rho) \geq cmv \geq cm^{-1/2}$. Moreover

$$s = m(1 - t) + (t - p_1) - \sum_{i=2}^m e_i$$

lies in $[1, 100]$ for large m , since $|t - p_1| \leq 1$. Exact permutation fairness and uniform face probabilities come from the realization theorem. $\square$

Matching upper bound. The subsequently proved theorem in `independent_directions/natural_scale_endpoint_profile.tex` gives

$$\max_i |m(1 - p_i) - s| \leq C_L (\sqrt{s/m} + m^{-1}) \quad (0 \leq s \leq L)$$

for each fixed L , including arbitrary positive row weights. The present examples force the square-root order at fixed positive s ; `endpoint_profile_sharpness.tex` separately forces the inverse- m additive floor, even for actual finite uniform dice. The subsequent note `independent_directions/intermediate_endpoint_sharpness.tex` constructs matching finite examples at every bounded positive endpoint scale, with realized deficit comparable to the prescribed scale.